



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

ROS/ROS2 PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Robotics

TOTAL: 10 QUESTIONS

Q1 What is ROS/ROS2 and what problem in Robotics does it solve? Result Model

Q6 How does ROS/ROS2 handle reliability and high availability? Reliability

Q2 What are the core components or concepts in ROS/ROS2? Lecture

Q7 What observability does ROS/ROS2 provide out of the box? Cycle

Q3 How does ROS/ROS2 compare to its main alternatives? Configuration

Q8 What are common performance bottlenecks in ROS/ROS2? Performance

Q4 How do you get started with ROS/ROS2 for a new project? Hardening

Q9 What security best practices apply when running ROS/ROS2? Securing

Q5 What configuration options most affect ROS/ROS2's behavior in production? Tuning

Q10 What mistakes do teams commonly make when adopting ROS/ROS2?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 ROS/ROS2 is a tool or platform that

Mitigation

ROS/ROS2 is a tool or platform that automates or simplifies a specific concern in the Robotics domain, reducing manual effort and operational complexity for engineering teams.

A6 ROS/ROS2 provides HA through leader election, replication, or stateless

horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 ROS/ROS2 has key building blocks that work

Primitives

ROS/ROS2 has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 ROS/ROS2 exposes metrics (Prometheus endpoint or built-in dashboard),

structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 ROS/ROS2 trades off simplicity against feature richness

Hardening

ROS/ROS2 trades off simplicity against feature richness differently than its alternatives. Choose ROS/ROS2 when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools,

large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install ROS/ROS2 via its package manager or binary release.

Gotchas

Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run ROS/ROS2 with the principle of least privilege

Assessment

Run ROS/ROS2 with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values,

Concurrency

retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial

and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.